

Familial Visceral Neuropathy 2 (VSCN2): Comprehensive Disease Characterization Report

Disease: Familial Visceral Neuropathy 2 (VSCN2) **OMIM:** #619465 · **MONDO:** MONDO:0030399 · **Gene:** *ERBB2* (HER2, 17q12; MIM 164870; HGNC:3430) **Category:** Mendelian (autosomal recessive neurocristopathy) **Report date:** 2026-08-31

Summary

Familial Visceral Neuropathy 2 (VSCN2; OMIM #619465) is an **ultra-rare autosomal recessive developmental neurocristopathy** caused by **biallelic loss-of-function variants in *ERBB2*** (HER2), the ligand-orphan but kinase-active partner of the neuregulin-1 (NRG1) receptor system. The disorder was defined in a single consanguineous Turkish family (a sister and brother) reported by Le et al. in 2021, who carried a homozygous kinase-domain missense variant, c.2129C>T; p.Ala710Val (A710V) in *ERBB2* (NM_004448.3). Functional assays showed this variant drastically reduces phosphorylation of both *ERBB2* and its heterodimeric partner *ERBB3*, establishing loss of NRG1/*ERBB* signaling as the mechanistic root of disease [PMID: 33497358].

Because *ERBB3* binds NRG1 with high affinity but is catalytically weak, and *ERBB2* is a ligand orphan but a strong kinase, functional NRG1 signaling requires an ***ERBB3/ERBB2* heterodimer**. Loss of *ERBB2* kinase activity therefore cripples signaling that is essential for the survival, proliferation, and migration of **neural-crest-derived enteric neuronal progenitors and Schwann cell precursors** that colonize the gut. The clinical consequence is a gastrointestinal dysmotility syndrome spanning colonic **aganglionosis/hypoganglionosis** (Hirschsprung disease, HSCR) and **chronic intestinal pseudo-obstruction (CIPO)**, accompanied by **peripheral axonal neuropathy, bilateral facial paresis, sensorineural hearing loss, ptosis**, and additional musculoskeletal and endocrine features. Gut histology in affected individuals revealed aganglionosis, hypoganglionosis, and intestinal smooth-muscle abnormalities [PMID: 33497358].

VSCN2 sits within an **allelic/pathway series** built around the NRG1/ERBB axis: VSCN1 (OMIM #243180, *ERBB3*), VSCN2 (OMIM #619465, *ERBB2*), and the dominant VSCN3 (#609629); *ERBB3* also underlies Lethal Congenital Contractural Syndrome type 2 (LCCS2, #607598). As of 2026, VSCN2 remains a **single-family entity** with a gene–disease validity classification of LIMITED, and there is **no disease-specific therapy** — management is supportive and surgical (pull-through for aganglionosis, nutritional support for dysmotility) with genetic counseling for prevention.

1. Disease Information

Overview. VSCN2 is a congenital disorder of gastrointestinal motility caused by defective development of the enteric nervous system (ENS) and peripheral nervous system, arising from biallelic *ERBB2* dysfunction. It is a **neurocristopathy** — a disease of neural-crest-derived tissues — combining features of Hirschsprung disease, chronic intestinal pseudo-obstruction, and peripheral neuropathy [PMID: 33497358;

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Key identifiers.

Resource	Identifier
OMIM	#619465 — <i>Visceral neuropathy, familial, 2, autosomal recessive</i>
MONDO	MONDO:0030399
Gene (OMIM)	<i>ERBB2</i> — MIM 164870
HGNC	HGNC:3430 (<i>ERBB2</i>)
MedGen	Concept present for the disorder
Orphanet	No dedicated ORPHA code specific to type 2 (VSCN1 = ORPHA99811)
ICD-11	No type-2-specific code (VSCN1 maps to DA90.2)

Synonyms / alternative names. Visceral neuropathy, familial, 2, autosomal recessive; VSCN2. Gene aliases for *ERBB2*: HER2, NEU, CD340, HER-2, VSCN2.

Data source type. Information is derived from **aggregated disease-level resources** (OMIM, MONDO, GenCC, PanelApp) and a **primary case series of individual patients** (Le et al. 2021), not EHR-scale populations. The entire VSCN2 clinical picture derives from **two siblings** in one consanguineous Turkish family.

2. Etiology

Disease causal factors. VSCN2 is a **monogenic, autosomal recessive genetic disorder**. The primary cause is **biallelic loss-of-function of *ERBB2***, disrupting NRG1/ERBB signaling required for ENS and Schwann cell development. There is no infectious or environmental cause; it is a developmental (neurocristopathy) mechanism.

*"Trio-exome sequencing led to the identification of biallelic variants in *ERBB3* and *ERBB2* in 8 individuals variably associating HSCR, CIPO, peripheral neuropathy, and arthrogryposis."*
[PMID: 33497358]

Genetic risk factors. The sole established genetic risk factor is inheritance of **two loss-of-function *ERBB2* alleles**. In the index family, the causal variant is homozygous p.Ala710Val, a kinase-domain missense change absent from gnomAD. **Consanguinity** is a major contributing factor (parents were heterozygous carriers).

Environmental risk factors. None established. As a fully penetrant Mendelian developmental disorder in the reported family, environmental exposures are not implicated.

Protective factors. None reported. No protective alleles or environmental modifiers are known. (Given a single family, statistical assessment of modifiers is not possible.)

Gene–environment interactions. None documented for VSCN2. Mechanistically, the disorder is driven by developmental loss of a receptor tyrosine kinase pathway rather than gene–environment interplay.

3. Phenotypes

The full phenotype spectrum derives from the two Turkish siblings (Le et al. 2021, family 5) plus the broader ERBB3/ERBB2 case series. Onset is **congenital/pediatric**, and expressivity is **variable** (HP:0003812).

Phenotype	Type	HPO term	Notes / frequency
Intestinal dysmotility / severe constipation	Clinical sign	HP:0002019 (constipation)	Core feature; irregular bowel evacuation
Colonic aganglionosis (Hirschsprung disease)	Pathology	HP:0002251 (aganglionic megacolon)	Rectal aganglionosis with submucosal nerve-fiber hyperplasia
Hypoganglionosis (above aganglionic segment)	Pathology	—	Present in gut histology
Chronic intestinal pseudo-obstruction (CIPO)	Clinical sign	HP:0002580	Neurogenic dysmotility
Peripheral axonal neuropathy	Clinical sign	HP:0003477 (peripheral neuropathy)	Axonal type
Bilateral facial paresis	Clinical sign	HP:0007209	Cranial nerve involvement
Sensorineural hearing loss	Lab/clinical	HP:0000407	—
Ptosis (unilateral)	Physical	HP:0007687 / HP:0000508	—
Hypotonia	Clinical sign	HP:0001252	—
Mild developmental delay	Behavioral/ neuro	HP:0001263	—
Clubfeet / talipes	Physical	HP:0001762	—
Scoliosis	Physical	HP:0002650	—
Micropenis	Physical	HP:0000054	Endocrine involvement
Low testosterone	Lab abnormality	HP:0040171	—
Small nose / hypoplastic alae nasi	Physical	HP:0000160 / HP:0000389	Facial dysmorphism
Arthrogryposis (broader series)	Physical	HP:0002804	Seen in ERBB3 cases; part of NRG1/ERBB spectrum

"biallelic variants in ERBB3 and ERBB2 in 8 individuals variably associating HSCR, CIPO, peripheral neuropathy, and arthrogryposis. Thorough gut histology revealed aganglionosis, hypoganglionosis, and intestinal smooth muscle abnormalities." [PMID: 33497358]

Severity/progression. Severe, congenital-onset, chronic. GI dysmotility is life-limiting and requires lifelong management. **Quality-of-life impact:** major — chronic constipation/pseudo-obstruction, dependence on nutritional support/surgery, hearing and neuromuscular impairment. No formal EQ-5D/SF-36 data exist for this ultra-rare entity.

4. Genetic / Molecular Information

Causal gene. *ERBB2* (HER2/NEU; 17q12; MIM 164870; HGNC:3430) — encodes a transmembrane EGF-family receptor tyrosine kinase.

Pathogenic variant (index family).

Feature	Detail
Variant	c.2129C>T (NM_004448.3), p.Ala710Val (A710V)
Zygosity	Homozygous (consanguineous; parents heterozygous)
Location	Highly conserved residue within the tyrosine kinase domain
Type	Missense
Population frequency	Absent from gnomAD
Functional consequence	Loss of function — drastic decrease in phosphorylation of both ERBB2 and ERBB3 (Neuro-2a overexpression)
Origin	Germline
ACMG interpretation	Pathogenic / likely pathogenic in context (segregation + functional + absence from population)

"The consequences of the identified variants were evaluated using quantitative real-time PCR (RT-qPCR) on patient-derived fibroblasts or immunoblot assays on Neuro-2a cells overexpressing WT or mutant proteins, revealing either decreased expression or altered phosphorylation of the mutant receptors." [PMID: 33497358]

Functional class. Loss of function (reduced receptor phosphorylation/kinase output). No gain-of-function or dominant-negative mechanism in VSCN2 (in contrast to oncogenic *ERBB2* amplification/activating mutations in cancer).

Related disorders / allelic series. - **VSCN1** — OMIM #243180, *ERBB3* (MIM 190151), autosomal recessive. - **VSCN2** — OMIM #619465, *ERBB2*, autosomal recessive. - **VSCN3** — OMIM #609629, autosomal dominant. - **LCCS2** (Lethal Congenital Contractural Syndrome 2) — OMIM #607598, *ERBB3*.

Modifier genes / epigenetics / chromosomal abnormalities. None specifically established for VSCN2. *SOX10* directly activates *ERBB3* transcription via an intronic neural-crest enhancer (*ERBB3_MCS6*) — relevant to the shared regulatory network but not a documented VSCN2 modifier. No large-scale chromosomal abnormalities are implicated.

5. Environmental Information

Environmental factors: None known. **Lifestyle factors:** None applicable. **Infectious agents:** None — VSCN2 is a genetic developmental disorder with no infectious trigger. This section is **not applicable** to VSCN2 beyond noting that consanguinity (a demographic, not environmental, factor) increases recessive risk.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *ERBB2* loss-of-function variant (p.A710V, kinase domain) → leads to a catalytically impaired *ERBB2* receptor with drastically reduced autophosphorylation/trans-phosphorylation capacity [PMID: 33497358].**

2. Because ERBB3 is ligand-binding but kinase-dead and ERBB2 is the kinase-active partner; impaired ERBB2 → **results in** failure of the **NRG1 → ERBB3/ERBB2 heterodimer** to trans-phosphorylate ERBB3 C-terminal tyrosines (Y1054–Y1328) [Reactome R-HSA-1227986].
3. Loss of ERBB3 phosphotyrosine docking sites → **results in** attenuated recruitment/activation of two downstream cascades (branch point):
4. **3a. PI3K → AKT survival arm** → normally phosphorylates/inhibits pro-apoptotic BAD, GSK3, FKHR-L1; its attenuation → **leads to** reduced survival of Schwann cell precursors and enteric progenitors [PMID: 11312610].
5. **3b. RAS → RAF → MEK → ERK (MAPK) proliferation/migration arm** → its attenuation → **leads to** impaired proliferation and migration of progenitors.
6. Reduced survival/proliferation/migration of **neural-crest-derived enteric neuronal progenitors and Schwann cell precursors (SCPs)** → **results in** failure of these cells to fully colonize the gut and populate peripheral nerves [PMID: 33720042].
7. Incomplete ENS colonization → **results in colonic aganglionosis (rectum) and hypoganglionosis** above the aganglionic segment, with compensatory submucosal nerve-fiber hyperplasia [PMID: 33497358].
8. Absent/atrophic enteric ganglia and smooth-muscle abnormalities → **lead to** loss of coordinated peristalsis → clinical **Hirschsprung disease / chronic intestinal pseudo-obstruction** (functional obstruction, severe constipation).
9. In parallel (branch), SCP/Schwann-cell deficiency along peripheral and cranial nerves → **leads to peripheral axonal neuropathy, bilateral facial paresis, sensorineural hearing loss, and ptosis**; neural-crest contributions to other structures → musculoskeletal/craniofacial and endocrine features (*inferred* from neural-crest biology and the broader ERBB3/ERBB2 series).

Detail by category

- **Molecular pathways:** NRG1/ERBB receptor tyrosine kinase signaling → PI3K–AKT and RAS–MAPK (KEGG "ErbB signaling pathway"; Reactome "Signaling by ERBB2", R-HSA-1227986). cAMP/PKA potentiates NRG1-induced ERBB2/ERBB3 phosphorylation and both downstream arms.
- **Cellular processes:** Progenitor survival (anti-apoptosis via AKT→BAD), proliferation, and directed migration (via GRB2/SHC, PLCγ/Ca²⁺-calcineurin, FAK, Rac/Cdc42).

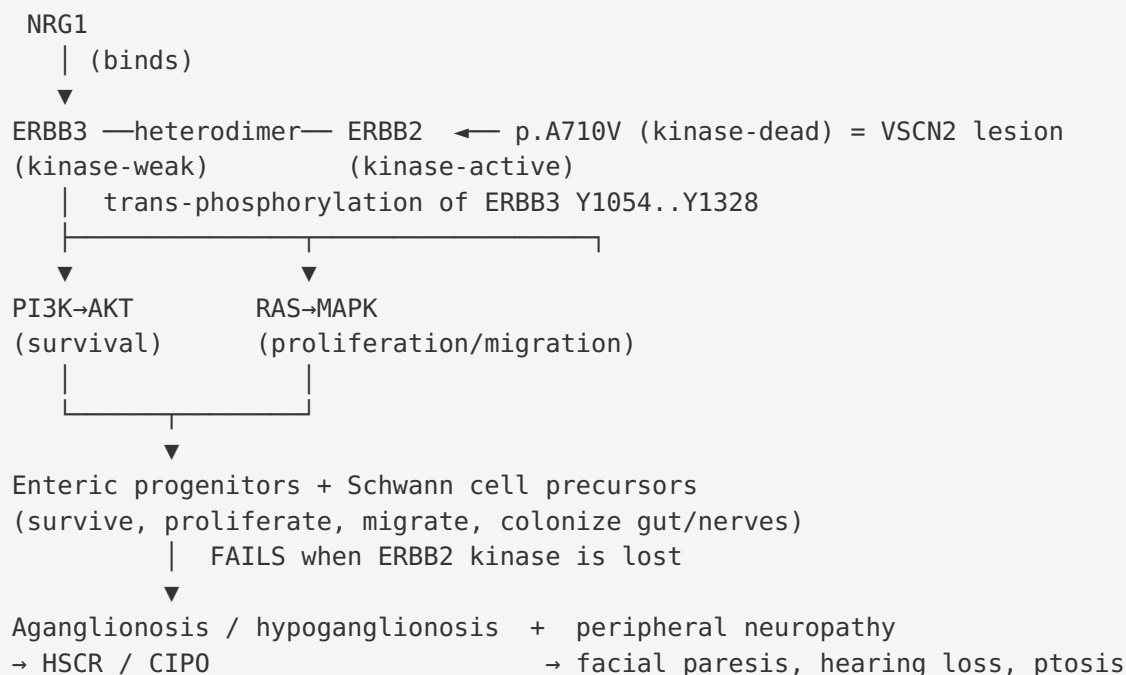
- **Protein dysfunction:** Loss of ERBB2 kinase output; the receptor pair is **asymmetric and complementary** — ERBB3 (high-affinity NRG1 binding, weak kinase) + ERBB2 (ligand-orphan, active kinase). A710V abolishes the kinase contribution.
- **Immune involvement:** Not implicated.
- **Tissue damage mechanism:** Developmental failure/hypoplasia of neural-crest derivatives (not oxidative/ischemic injury).
- **Molecular profiling:** Mouse single-cell RNA-seq confirmed *ErbB3/ErbB2* expression in **enteric neuronal progenitor cells**; a conditional *ErbB3*-deficient model revealed a primary role for ERBB3 in enteric progenitors [PMID: 33497358;

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"Experiments using mice revealed that *ErbB3* and *ErbB2* were expressed in enteric neuronal progenitor cells." [PMID: 33720042]

"The cell type-specific *ErbB3* and *ErbB2* function was further analyzed in mouse single-cell RNA sequencing data and in a conditional *ErbB3*-deficient mouse model, revealing a primary role for ERBB3 in enteric progenitors." [PMID: 33497358]

ASCII schematic



GO term suggestions: GO:0038128 (ERBB2 signaling pathway), GO:0007399 (nervous system development), GO:0048484 (enteric nervous system development), GO:0014010 (Schwann cell proliferation), GO:0043524 (negative regulation of neuron apoptotic process), GO:0016477 (cell migration). **CL term suggestions:** CL:0011103 (enteric neuron), CL:0002375 (Schwann cell precursor), CL:0000333 (migratory neural crest cell).

7. Anatomical Structures Affected

Organ level (primary): Large intestine / colon and rectum (UBERON:0001155 colon; UBERON:0001052 rectum) — site of aganglionosis. Small and large bowel broadly in CIPO. **Secondary / body systems:** Digestive system (UBERON:0001007), peripheral nervous system (UBERON:0000010), cranial nerves (facial nerve UBERON:0001647), inner ear/cochlea (sensorineural hearing loss; UBERON:0001846), musculoskeletal (spine, feet), and reproductive/endocrine (micropenis, low testosterone).

Tissue/cell level: Enteric ganglia within the myenteric (Auerbach; UBERON:0002439) and submucosal (Meissner; UBERON:0013529) plexuses; intestinal smooth muscle (UBERON:0001529); peripheral nerve Schwann cells. Cell populations: **enteric neurons and their progenitors (CL:0011103), Schwann cells / SCPs (CL:0002573 / CL:0002375)**, neural-crest-derived cells.

Subcellular level: Plasma membrane receptor tyrosine kinase complex (GO:0005886 plasma membrane; GO:0004714 transmembrane receptor protein tyrosine kinase activity). No mitochondrial/lysosomal defect implicated.

Localization / lateralization: GI involvement follows the aganglionic segment (distal colon/rectum). Facial paresis is **bilateral**; ptosis reported **unilateral**; overall a mixed lateralization pattern.

8. Temporal Development

- **Onset:** Congenital / neonatal–pediatric. GI dysmotility and HSCR features manifest early in life; developmental/craniofacial and neuromuscular features are present from birth.
- **Onset pattern:** Chronic, developmental (insidious progression of dysmotility complications).

- **Progression:** Chronic and lifelong. Enteric aganglionosis is a fixed developmental deficit; secondary complications (pseudo-obstruction, nutritional failure) can be progressive without intervention.
 - **Course pattern:** Chronic-progressive with episodic pseudo-obstruction crises; not relapsing-remitting.
 - **Remission:** No spontaneous remission; surgical management (pull-through) can relieve the aganglionic obstruction but does not cure the underlying neuropathy.
 - **Critical periods:** The developmental window of neural-crest colonization of the gut (embryonic) is the mechanistically critical period — clinically not therapeutically accessible postnatally.
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9. Inheritance and Population

- **Inheritance: Autosomal recessive.** Homozygous causal variant in the consanguineous index family; heterozygous parents unaffected.
- **Penetrance:** Presumed complete in biallelic carriers (based on single family; formal penetrance estimate not possible).
- **Expressivity:** Variable across the broader NRG1/ERBB series (HP:0003812).
- **Consanguinity:** Central — the disorder was identified in a consanguineous Turkish family; consanguinity is the main route to homozygosity.
- **Founder effect / carrier frequency:** Not established; variant absent from gnomAD, so no population carrier frequency is defined.
- **Epidemiology: Ultra-rare** — VSCN2 remains a **single-family entity** as of 2026, with no additional ERBB2-specific families reported (2022–2026 literature review). No prevalence/incidence figures exist. Context: Hirschsprung disease overall has an incidence of ~1 in 5,000 live births, but VSCN2 is a vanishingly small molecular subset.
- **Demographics:** Reported in a Turkish family (one affected sister and one affected brother); sex ratio not meaningfully estimable (both sexes affected). Gene–disease validity: **LIMITED** (GenCC/PanelApp).

"The patients carried homozygous or heterozygous variants in ERBB3 or ERBB2, which encode transmembrane epidermal growth factor receptors that bind neuroregulin 1 (NRG1)."
[PMID: 33720042]

10. Diagnostics

Molecular diagnosis (definitive): - **Whole-exome sequencing (WES)** — the method that established the diagnosis (trio/WES in Le et al. 2021). Trio design aids interpretation. - **Whole-genome sequencing (WGS)** — useful for non-coding/structural variants when WES is negative. - **Targeted NGS gene panels** for HSCR/CIPO/pediatric intestinal pseudo-obstruction (e.g., **Genomics England PanelApp panel 1217 includes ERBB2**). - **Single-gene ERBB2 sequencing with segregation** (confirm parental heterozygosity).

"Trio-exome sequencing led to the identification of biallelic variants in ERBB3 and ERBB2."
[PMID: 33497358]

Histopathology (cornerstone for the aganglionosis component): - **Rectal suction biopsy** — absence of ganglion cells with hypertrophied acetylcholinesterase (AChE)-positive cholinergic nerve trunks. - **Calretinin immunohistochemistry** — loss of staining is characteristic of Hirschsprung disease (sensitivity ~80–100%, specificity ~99–100%). - Full-thickness bowel biopsy in selected cases; findings include aganglionosis, hypoganglionosis, submucosal nerve-fiber hyperplasia, and smooth-muscle abnormalities.

CIPO workup: Exclude mechanical obstruction (CT/MRI); metabolic panel; endoscopy; GI scintigraphy; **antroduodenal/small-bowel manometry** (neuropathic pattern — abnormal migrating motor complex, non-propagated bursts); **anorectal manometry** (absent recto-anal inhibitory reflex in HSCR).

Adjunct testing: Nerve conduction studies/EMG (confirm axonal peripheral neuropathy); audiometry (sensorineural hearing loss); ophthalmologic assessment (ptosis).

Differential diagnosis: VSCN1 (*ERBB3*) and VSCN3; syndromic HSCR (e.g., *RET*, *EDNRB*, *SOX10* — Waardenburg-Shah); other causes of CIPO (myogenic vs neurogenic); mitochondrial neurogastrointestinal encephalomyopathy (MNGIE); LCCS2. Molecular testing distinguishes these.

Screening: Cascade carrier testing in the family; prenatal/preimplantation testing feasible once the familial variant is known.

11. Outcome / Prognosis

- **Survival / mortality:** No formal survival statistics for this ultra-rare entity. Prognosis is driven by the severity of intestinal dysmotility; CIPO with intestinal failure carries substantial morbidity/mortality and may require long-term parenteral nutrition.
 - **Morbidity / function:** High chronic morbidity — dependence on nutritional support and/or surgery, peripheral neuropathy, hearing impairment, and developmental delay affecting daily functioning.
 - **Complications:** Pseudo-obstruction crises, malnutrition/intestinal failure, enterocolitis (HSCR-associated), complications of surgery and long-term nutrition support.
 - **Recovery potential:** The developmental enteric deficit is not reversible; surgery relieves the aganglionic segment but underlying dysmotility and extraintestinal features persist.
 - **Prognostic factors:** Extent of aganglionosis/dysmotility, presence of intestinal failure, and severity of extraintestinal (neuromuscular, hearing) involvement. No validated molecular prognostic biomarkers.
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12. Treatment

There is no disease-specific pharmacotherapy, gene therapy, or cell therapy for VSCN2. Management is **supportive and surgical**, targeting the manifestations.

Modality	Intervention	NCIT suggestion
Surgical (HSCR)	Pull-through resection of aganglionic segment (Soave, Duhamel, Swenson)	NCIT:C157763 (surgical procedure)
Nutritional	Enteral/parenteral nutrition for dysmotility/intestinal failure	NCIT:C15184 (nutritional support)
GI supportive	Prokinetics, laxatives, bowel decompression for CIPO	NCIT:C1516 (supportive care)
Neuromuscular	Physical/occupational therapy for neuropathy, hypotonia	NCIT:C15315 (rehabilitation therapy)
Sensory	Hearing aids / audiologic support for sensorineural hearing loss	—
Endocrine	Testosterone/androgen evaluation for micropenis/low testosterone	—
Ophthalmology	Ptosis management	—

- **Pharmacogenomics:** Not applicable (no targeted drug).
- **Advanced / experimental therapeutics:** No gene, RNA, or cell therapies in trials for VSCN2. (The many *ERBB2*/HER2-targeted oncology agents — e.g., trastuzumab, trastuzumab deruxtecan — are irrelevant here, as VSCN2 is caused by loss, not gain, of *ERBB2* function; HER2 antagonism would be counterproductive.)
- **Treatment strategy:** Multidisciplinary care (pediatric surgery, gastroenterology/motility, neurology, audiology, nutrition, genetics). Personalized approach centered on the individual's dysmotility burden.

13. Prevention

- **Primary prevention: Genetic counseling** for consanguineous families and known carriers; awareness of autosomal recessive recurrence risk (25% for carrier couples).
- **Secondary prevention: Cascade carrier screening** within the affected family; **prenatal diagnosis or preimplantation genetic testing (PGT)** once the familial *ERBB2* variant is

identified. Early recognition of HSCR/CIPO in at-risk newborns enables timely surgical/nutritional intervention.

- **Tertiary prevention:** Prevention of complications — enterocolitis prophylaxis in HSCR, nutritional monitoring to prevent intestinal failure, and rehabilitation to limit neuromuscular disability.
- **Immunization / public-health / environmental interventions:** Not applicable (non-infectious genetic disorder).
- **Counseling:** NSGC/ACMG-guided genetic counseling is the principal preventive tool.

14. Other Species / Natural Disease

- **Taxonomy / model species:** *Mus musculus* (NCBI:txid10090) and *Danio rerio* (zebrafish, NCBI:txid7955) are the principal experimental species; no naturally occurring companion-animal or wildlife VSCN2 analog is catalogued in OMIA.
- **Orthologous genes:** Mouse *ErbB2* (NCBI Gene 13866), *ErbB3* (NCBI Gene 13867), *Nrg1* (NCBI Gene 211323); strongly conserved.
- **Comparative biology / conservation:** The NRG1/ERBB3/ERBB2 axis is deeply conserved across vertebrates in Schwann cell and neural-crest development. Mouse and zebrafish loss-of-function phenocopy key components (Schwann cell precursor loss; reduced enteric/sympathetic ganglia).
- **Zoonotic potential:** None (genetic disorder).

15. Model Organisms

Mouse and zebrafish models robustly validate the ERBB2/ERBB3/NGR1 mechanism.

Model	Phenotype	Recapitulation of VSCN2	Reference
<i>Erb2</i> -null mouse	Embryonic lethal ~E10.5 (cardiac ventricular trabeculation defect)	Limited by early lethality; ENS not fully assessable	Erickson 1997 [PMID: 9362461]
<i>Nrg1</i> (heregulin)-null mouse	Embryonic lethal ~E10.5 (cardiac)	Confirms NRG1 as the ligand of the axis	[PMID: 9362461]
<i>Erb3</i> -null mouse	Lethal ~E13.5; absent Schwann cell precursors ; generalized neural-crest defect (reduced cranial ganglia, enteric ganglia, adrenal chromaffin cells)	Strong — recapitulates peripheral neuropathy + reduced enteric ganglia	Riethmacher 1997 [PMID: 9338783]
Cardiac-rescued <i>Erb2</i> -null mouse	Survive to birth; lack Schwann cells , lose motor/sensory neurons	Strong for peripheral component	Woldeyesus/Morris, Neuron 1999
<i>Wnt1-Cre;Erb3</i> conditional	Depletes cervical sympatho-enteric population (E10); esophageal ganglia reduced ~68% at E13.5; reduced foregut Sox10+ cells at E10.5	Directly models enteric progenitor failure	[PMID: 33497358]
<i>erb2/erb3</i> mutant zebrafish	Early ENS grossly normal (redundancy/timing; dual vagal SCP + trunk crest origin)	Highlights model limitation/redundancy	(Le et al. discussion)

"Schwann-cell precursors" — Riethmacher et al. showed *Erb3*-null mice lack Schwann-cell precursors, recapitulating the peripheral neuropathy component. [PMID: 9338783]

Model limitations: Conventional *Erb2/Nrg1* knockouts die too early (cardiac lethality) to assess ENS directly; conditional/rescue models are required. Zebrafish show ENS redundancy, so early enteric formation can appear normal. These caveats reflect pathway redundancy and the dual embryonic origin of enteric neurons rather than absence of a role.

Applications: Single-cell RNA-seq and conditional deletion models establish cell-type-specific ERBB function in enteric progenitors and Schwann cell precursors — the core disease mechanism.

Mechanistic Model / Interpretation

VSCN2 is best understood as a **kinase-deficiency neurocristopathy**. The NRG1 receptor system is intrinsically split into a ligand-binding subunit (ERBB3) and a catalytic subunit (ERBB2). Neither works alone: signaling demands the **ERBB3/ERBB2 heterodimer**, in which ERBB2 phosphorylates ERBB3's cytoplasmic tail to create docking sites for PI3K–AKT (survival) and RAS–MAPK (proliferation/migration). The VSCN2 lesion (homozygous p.A710V in the ERBB2 kinase domain) removes the catalytic half of this obligate partnership, functionally silencing NRG1 signaling despite intact ligand and intact ERBB3.

Developmentally, the cells most dependent on this signal are **neural-crest-derived enteric progenitors and Schwann cell precursors**, which must survive, proliferate, and migrate to colonize the gut and peripheral nerves. Their signaling failure produces the disease's dual signature: **enteric aganglionosis/hypoganglionosis** (→ HSCR/CIPO) in the gut and **Schwann-cell/peripheral-nerve deficiency** (→ axonal neuropathy, facial paresis, sensorineural hearing loss, ptosis). Mouse genetics corroborates each branch — *ErbB3/ErbB2/Nrg1* nulls lose Schwann cell precursors and show reduced enteric ganglia, and conditional *Wnt1-Cre;ErbB3* deletion specifically depletes sympatho-enteric progenitors. The convergence of human genetics, patient-cell functional assays (reduced ERBB2/ERBB3 phosphorylation), single-cell expression data, and conditional mouse models makes this one of the more mechanistically complete rare-disease causal chains.

The disorder anchors a coherent **allelic/pathway series**: recessive *ERBB3* (VSCN1, LCCS2) and recessive *ERBB2* (VSCN2) lesions produce overlapping neurocristopathy phenotypes, consistent with their shared obligate-heterodimer biology.

Evidence Base

PMID	Title / focus	Role in this report
33497358	<i>Dysregulation of the NRG1/ERBB pathway causes a developmental disorder with gastrointestinal dysmotility in humans</i> (Le et al., J Clin Invest 2021)	Primary defining paper. Identifies biallelic <i>ERBB2/ERBB3</i> variants in 8 individuals; defines VSCN2 (<i>ERBB2</i> A710V); functional loss-of-function assays; single-cell + conditional mouse validation.
33720042	<i>Hirschsprung disease and more: dysregulation of ERBB2 and ERBB3</i> (Gershon commentary)	Confirms <i>ERBB2/ERBB3</i> as NRG1-binding EGF receptors expressed in enteric neuronal progenitors; frames HSCR/CIPO as congenital motility defects.
9338783	Riethmacher et al. 1997 — <i>ErbB3</i> -null mouse	Model evidence: absence of Schwann-cell precursors; neural-crest/enteric ganglia defect.
9362461	Erickson et al. 1997 — <i>ErbB2/Nrg1</i> nulls	Model evidence: early cardiac-lethal phenotypes; establishes ligand/receptor pairing.
11312610	<i>Neuregulin signaling through a PI3K/Akt/Bad pathway in Schwann cell survival</i>	Mechanistic: defines the PI3K/AKT/BAD survival arm downstream of NRG1-ERBB.

Evidence-source	types:	Human	clinical/genetic
(P 33497358 commentary 33720042); (P 9338783 P 9362461 conditional models); in vitro functional (patient fibroblast RT-qPCR, Neuro-2a immunoblot; P 33497358);			organism

biochemical/pathway

(P 11312610;
Reactome R-HSA-1227986).

(Note: The breast-cancer HER2 therapy papers retrieved during investigation — PMIDs 40759100, 32715420, 28366406, 18650157 — pertain to oncologic HER2 gain-of-function targeting and are not applicable to VSCN2, a loss-of-function disorder. They are excluded from the mechanistic evidence base.)

Limitations and Knowledge Gaps

1. **Single-family disease.** All VSCN2-specific clinical data derive from two siblings in one consanguineous Turkish family. Penetrance, expressivity, natural history, prognosis, and prevalence cannot be estimated; gene–disease validity is **LIMITED**.
2. **One variant.** Only p.A710V is documented for *ERBB2* in VSCN2; the full mutational spectrum, genotype–phenotype correlations, and allele frequencies are unknown.
3. **No epidemiologic data.** No prevalence/incidence, sex ratio, or geographic distribution beyond the index Turkish family.
4. **No dedicated Orphanet/ICD-11 code** for type 2, complicating standardized cataloguing.
5. **Model caveats.** Conventional *ErbB2* knockouts are cardiac-lethal before ENS assessment; zebrafish show ENS redundancy — so the enteric role relies on conditional/rescue models and human genetics rather than a direct *ErbB2* enteric knockout phenotype.
6. **No therapeutics.** No disease-specific treatment exists; management is empiric/supportive.
7. **Extraintestinal features under-characterized.** The mechanistic link between *ERBB2* loss and craniofacial/endocrine features (micropenis, low testosterone, hypoplastic alae nasi) is *inferred* from neural-crest biology, not directly demonstrated.

Proposed Follow-up Experiments / Actions

1. **Patient ascertainment / matchmaking.** Deposit the *ERBB2* variant in ClinVar and use GeneMatcher/Matchmaker Exchange to find additional VSCN2 families, enabling genotype–phenotype and penetrance analysis.

2. **Allelic-series functional platform.** Establish a standardized ERBB2/ERBB3 phosphorylation and PI3K-AKT/MAPK reporter assay to classify future *ERBB2* variants (VUS resolution per ACMG PS3/BS3).
 3. **Conditional *ErbB2* enteric knockout.** Generate a *Wnt1-Cre;ErbB2* (or inducible *Sox10-CreER;ErbB2*) mouse to directly test the enteric requirement for ERBB2 (paralleling the existing *ErbB3* conditional), bypassing cardiac lethality.
 4. **Patient iPSC → enteric neural crest / organoid model.** Differentiate patient-derived iPSCs into enteric neural crest and gut organoids to model colonization failure and screen small molecules acting downstream of the receptor block.
 5. **Extraintestinal mechanism mapping.** Lineage-tracing in conditional models to test whether craniofacial/endocrine features arise from specific neural-crest sublineages, formalizing the currently inferred steps.
 6. **Registry / natural history.** Create an NRG1/ERBB-spectrum (VSCN1/2/3, LCCS2) registry to aggregate ultra-rare cases and define prognosis and management outcomes.
 7. **Ontology curation.** Propose a dedicated Orphanet/ICD-11 entry and HPO annotation set for VSCN2 to improve knowledge-base interoperability.
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Prepared from a 5-iteration autonomous investigation. Core citations: [PMID 33497358](#), [PMID 33720042](#), [PMID 9338783](#), [PMID 9362461](#), [PMID 11312610](#).
